Spatially-aware deep learning for high-resolution snow depth mapping in a Pyrenean catchment: evaluating pattern fidelity with the SPAEF metric

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Abstract

Accurate, high-resolution snow depth (HS) mapping is essential for water resource management, avalanche forecasting and ecological studies in mountain regions, yet remains challenging due to the strong spatial heterogeneity of the snowpack. We present a deep-learning framework that predicts HS at 1 m resolution in the Izas experimental catchment (Central Spanish Pyrenees) from a multi-channel stack of topographic, snow-cover and wind-redistribution predictors derived from a 27-date airborne LiDAR time series (2021–2025). We benchmark a Random Forest (RF) baseline against two convolutional architectures (U-Net and ResUNet++) and show that conventional pixel-wise metrics such as R^2 are insufficient to assess the quality of the predicted spatial patterns. We adopt the Spatial Efficiency metric (SPAEF) and a multi-scale variant (MSPAEF) as complementary, pattern-oriented evaluation criteria, and introduce a hybrid spatial loss that combines pixel-wise mean squared error with a spatial correlation term weighted by a hyperparameter λ. Across three random seeds, ResUNet++ trained with the spatial loss ($\lambda = 0.4$) achieves the best and most reproducible performance ($R^2 = 0.208 \pm 0.027$, SPAEF $= 0.297 \pm 0.018$), substantially outperforming RF (SPAEF $= 0.107$) and a markedly unstable U-Net (SPAEF $= 0.088 \pm 0.216$). A leave-one-group-out ablation identifies fine-to-coarse topography (5 m DEM derivatives) and snow persistence as the most informative predictors, whereas satellite snow-cover extent contributes marginally. Augmenting the input with

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meteorological forcing (temperature and accumulated precipitation) does not yield a statistically significant improvement, indicating that the topographic and snow-history signals already capture most of the explainable variance for this catchment. Our results highlight the importance of pattern-aware metrics and losses for high-resolution snow mapping and provide a reproducible benchmark for future work.

Keywords: snow depth, deep learning, ResUNet++, SPAEF, spatial loss, LiDAR, Pyrenees, Random Forest

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1. Introduction

Seasonal snow is a key component of the mountain water cycle. In Mediterranean mountain ranges such as the Pyrenees, snowmelt sustains river discharge, reservoir storage, hydropower generation and irrigation during the dry season, and the spatial distribution of the snowpack governs avalanche hazard, soil moisture and the timing of vegetation growth [1, 2]. Snow depth (HS) is highly variable in space: wind redistribution, terrain shading, slope, aspect and preferential deposition produce sharp gradients over distances of a few metres, especially in wind-exposed alpine terrain [3, 4]. Capturing this fine-scale variability is therefore essential, but it is precisely where coarse-resolution products and point measurements fail.

Airborne and terrestrial Light Detection and Ranging (LiDAR) surveys have become the reference technique for mapping HS at sub-metre resolution by differencing snow-on and snow-off digital surface models [5, 6]. However, LiDAR acquisitions are expensive and logistically demanding, so they can only be carried out on a limited number of dates per season. This motivates the development of predictive models that can *interpolate and extrapolate* HS from a small set of LiDAR surveys, using spatially exhaustive and inexpensive predictors such as terrain attributes, satellite-derived snow-cover information and meteorological forcing.

Two broad families of methods have been applied to this problem. Statistical and machine-learning regressors (multiple linear regression, generalised additive models, Random Forests, gradient boosting) operate on a per-pixel basis and have proven effective at relating HS to topographic predictors [4, 7]. Their main limitation is that they treat each pixel independently and therefore cannot exploit the spatial context of the snowpack. Convolutional neural networks (CNNs), by contrast, learn spatially structured representations and have recently been used for snow and related cryospheric mapping tasks [8, 9]. Yet most studies still evaluate model quality almost exclusively with pixel-wise metrics – coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE) – which reward the correct prediction of the mean and penalise outliers, but are largely insensitive to whether the model reproduces the *spatial pattern* of accumulation and scour. A model can attain a respectable R^2 while producing an over-smoothed map that misses the very drift features that matter for avalanche and hydrological applications.

In this work we address both the modelling and the evaluation sides of this gap. We build a high-resolution (1 m) HS prediction benchmark for the Izas experimental catchment in the Central Spanish Pyrenees, using a 27-date LiDAR time series (2021–2025) provided in collaboration with the Instituto Pirenaico de Ecología (IPE–CSIC). On top of this benchmark we (i) compare a Random Forest baseline with two CNN architectures (U-Net and ResUNet++) under a strict temporal train/validation/test split, (ii) adopt the Spatial Efficiency metric (SPAEF) and a multi-scale variant (MSPAEF) as pattern-oriented evaluation criteria alongside the conventional pixel-wise metrics, and (iii) introduce a hybrid spatial loss that explicitly optimises for spatial pattern fidelity through a tunable correlation term.

1.1. Contributions

The main contributions of this paper are:

1. **A reproducible 1 m snow-depth benchmark** for a Pyrenean catchment, built from airborne LiDAR and a rich stack of 22 topographic, snow-cover and wind-redistribution predictors, with a temporally disjoint test year (2025) that stresses the extrapolation ability of the models.
2. **A pattern-aware evaluation protocol** based on SPAEF and MSPAEF, which we show to be far more discriminative than R^2 for this task – in particular, it exposes the severe seed-to-seed instability of U-Net that a single-run R^2 would hide.
3. **A hybrid spatial loss** combining pixel-wise mean squared error with a spatial Pearson-correlation term weighted by a hyperparameter λ, together with a complete sweep ($\lambda \in [0, 1]$, eight values, three seeds) that identifies $\lambda = 0.4$ as the optimum, improving both R^2 and SPAEF while halving the variance across seeds.

4. **A leave-one-group-out ablation** of the predictor channels that quantifies the relative importance of fine/coarse topography, snow persistence, wind sheltering and satellite snow-cover extent.
5. **An analysis of meteorological forcing** (air temperature and accumulated precipitation as additional channels), showing that, for this catchment and predictor set, meteorological inputs do not significantly improve the prediction once topography and snow history are available.

1.2. Paper organisation

The remainder of the paper is organised as follows. Section 2 reviews related work on snow-depth modelling, deep learning for environmental mapping and spatial evaluation metrics. Section 3 describes the study area, the LiDAR-derived dataset and the predictor channels. Section 4 presents the models, the SPAEF/MSPAEF metrics and the spatial loss, together with the experimental protocol. Section 5 reports the model comparison, the λ sweep, the channel ablation and the meteorological experiments. Section 6 discusses the implications and limitations, and Section 7 concludes.

2. Related work

2.1. Snow-depth and SWE mapping from terrain and remote sensing

The spatial distribution of mountain snow has traditionally been modelled by relating snow depth or snow water equivalent (SWE) to terrain attributes through statistical regression. Elevation, slope, aspect (often decomposed into northness and eastness), curvature and topographic position indices are the classical predictors [10, 11, 4]. Wind redistribution, a dominant control in alpine terrain, is commonly represented by terrain-based parameters such as the maximum upwind slope S_x [12, 13], which we also adopt here. More recently, machine-learning regressors – particularly Random Forests [14] – have become a de-facto standard for HS/SWE interpolation because they handle non-linear interactions and large predictor sets robustly [7, 9, 15]. These pixel-wise methods are strong baselines but, by construction, ignore the spatial arrangement of neighbouring pixels.

2.2. Deep learning for high-resolution environmental mapping

Convolutional neural networks have transformed dense prediction tasks in remote sensing, from land-cover classification and change detection [8] to super-resolution and regression of continuous geophysical fields. Encoder–decoder architectures with skip connections, of which the U-Net [16] is the archetype, are especially well suited to mapping problems because they combine multi-scale context with spatially precise outputs. Residual and attention-augmented variants such as ResUNet [17] and ResUNet++ [18] add residual blocks, squeeze-and-excitation and atrous spatial pyramid pooling, which stabilise training and improve the modelling of multi-scale structures. In the snow context, CNNs and other deep models have been explored for snow-cover mapping, SWE reconstruction and downscaling [9, 19], but high-resolution (metre-scale) HS regression from LiDAR remains comparatively under-explored, and systematic comparisons against strong tabular baselines under temporally disjoint evaluation are scarce.

2.3. Evaluation metrics for spatial fields

Pixel-wise metrics (R^2, RMSE, MAE, bias, Nash–Sutcliffe efficiency) dominate the snow-modelling literature, yet they assess only the marginal agreement between predicted and observed values and are blind to the spatial organisation of the error. The Spatial Efficiency metric (SPAEF) introduced by **(author?)** [20] was designed precisely to evaluate the fidelity of spatial patterns, by combining three complementary components: the Pearson correlation between the two fields, the ratio of their coefficients of variation, and the overlap of their value histograms. SPAEF and related multi-component metrics have since been adopted to benchmark distributed hydrological and land-surface models [20, 21]. Despite their suitability, such pattern-oriented metrics are rarely used for high-resolution snow mapping. We argue, and demonstrate empirically, that they are necessary to obtain an honest picture of model quality and stability, and we further use them – through a differentiable spatial-correlation term – to *guide* the optimisation, not merely to evaluate it.

2.4. Pattern-oriented and physics-aware losses

Optimising solely a pixel-wise loss (MSE/MAE) tends to produce regression to the mean and over-smoothed fields. To counteract this, a growing body of work augments the training objective with structural or perceptual terms – for example structural similarity (SSIM) losses, gradient-difference losses, or adversarial terms – so that the model is rewarded for reproducing spatial structure [22]. In the environmental domain, correlation-based and multi-scale loss terms have been used to better preserve spatial texture.

Building on this idea, we propose a simple and interpretable hybrid loss that linearly blends MSE with a spatial Pearson-correlation penalty controlled by a single hyperparameter λ, and we characterise its effect on both accuracy and spatial pattern fidelity through a full sweep.

3. Study area and data

3.1. Study area

The study is carried out in the Izas experimental catchment, a small ($\sim 0.55\,\text{km}^2$) alpine headwater basin located in the Central Spanish Pyrenees (province of Huesca), at elevations of approximately 2000–2300 m a.s.l. The catchment has been monitored for decades by the Instituto Pirenaico de Ecología (IPE–CSIC) and is characterised by a strongly wind-influenced snowpack: persistent north-westerly winds generate pronounced snow redistribution, with deep drifts on lee slopes and scoured wind-exposed ridges. This produces the sharp, repeatable spatial patterns of accumulation that make Izas an ideal testbed for high-resolution HS prediction. The LiDAR and meteorological data used in this study were provided through a collaboration with IPE–CSIC.

3.2. LiDAR snow-depth surveys

Snow depth was derived from airborne LiDAR surveys by differencing snow-on digital surface models against a snow-off reference, yielding HS maps at 1 m spatial resolution. A total of 27 acquisition dates spanning four hydrological years (2021–2025) are available. The resulting HS rasters constitute the prediction target (ground truth) of all models. To enable convolutional processing, each dated HS map and its co-registered predictor stack are tiled into fixed-size patches; pixels outside the catchment mask or without valid HS are excluded from both training and evaluation.

3.3. Predictor channels

Each training sample is a multi-channel image in which the predictors are spatially co-registered with the target HS map at 1 m resolution. The base configuration comprises **22 channels**, which we group into four families (Table 1):

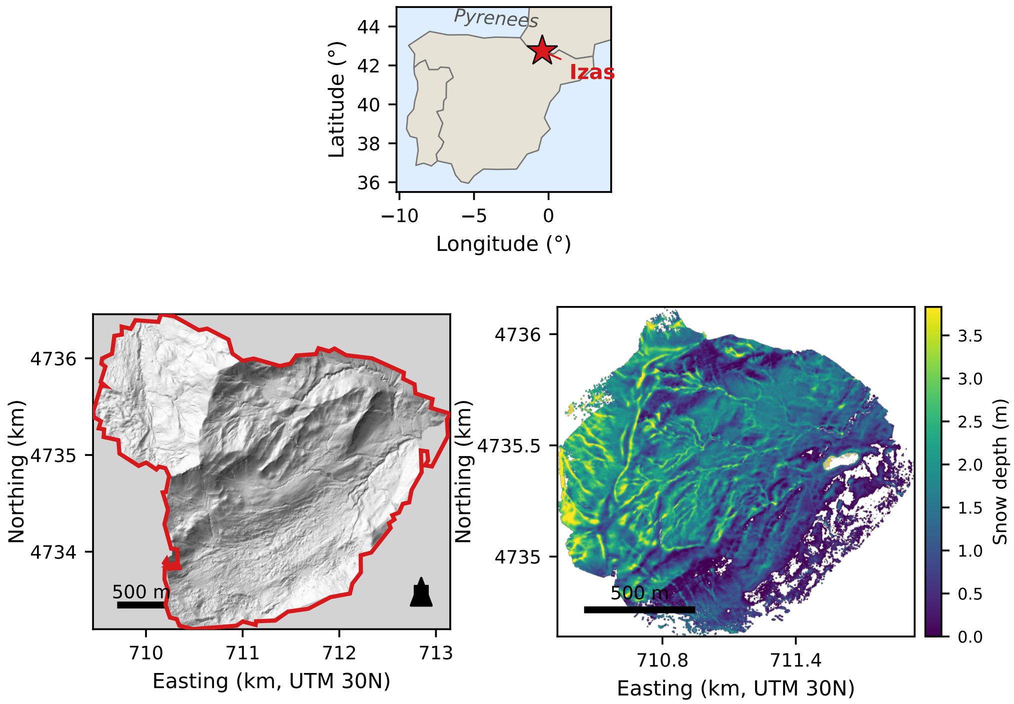

Figure 1: Location and physiography of the Izas experimental catchment (Central Spanish Pyrenees). (a) Regional setting within the Iberian Peninsula and the Pyrenees; (b) outline of the LiDAR-surveyed footprint (union of all acquisition dates) over a hillshade of the 1 m DEM (EPSG:25830, UTM zone 30N); (c) example LiDAR-derived snow-depth map (1 April 2025) illustrating the wind-driven accumulation patterns the models must reproduce.

- **Fine-scale topography (1 m).** Elevation (DEM), slope, northness (cos of aspect) and eastness (sin of aspect), and the Topographic Position Index (TPI). These describe the local terrain that controls radiation, gravitational transport and small-scale deposition.

- **Coarse-scale topography (5 m).** The same five terrain derivatives computed from a 5 m DEM. The coarser scale captures the meso-topographic context (ridges, gullies, broad lee slopes) that governs wind redistribution over longer fetch distances.

- **Wind sheltering.** The maximum upwind slope index S_x [12] computed over a 200 m search distance for eight azimuths (0°, 45°, ..., 315°). These eight channels encode directional exposure/sheltering and

Table 1: The 22 base predictor channels and the four additional meteorological channels (26-channel configuration). "Static" channels are constant in time; "dynamic" channels vary between acquisition dates.

Idx	Channel	Family	Type
0	DEM (1 m)	Topo 1 m	static
1	Slope (1 m)	Topo 1 m	static
2	Northness (1 m)	Topo 1 m	static
3	Eastness (1 m)	Topo 1 m	static
4	TPI (1 m)	Topo 1 m	static
5	SCE (snow-cover extent)	Snow	dynamic
6–13	S_x 200 m (8 azimuths)	Wind	static
14	Persistence 15 d	Snow	dynamic
15	Persistence 30 d	Snow	dynamic
16	Persistence 60 d	Snow	dynamic
17	DEM (5 m)	Topo 5 m	static
18	Slope (5 m)	Topo 5 m	static
19	Northness (5 m)	Topo 5 m	static
20	Eastness (5 m)	Topo 5 m	static
21	TPI (5 m)	Topo 5 m	static
Additional meteorological channels (26-ch configuration)			
22	Mean air temp. 7 d	Meteo	dynamic (scalar)
23	Mean air temp. 15 d	Meteo	dynamic (scalar)
24	Mean air temp. 30 d	Meteo	dynamic (scalar)
25	Accum. precip. from Oct	Meteo	dynamic (scalar)

are the primary terrain-based proxy for wind-driven accumulation and scour.

- **Snow-cover history.** The fractional snow-cover persistence over the 15, 30 and 60 days preceding each acquisition, derived from a binary satellite snow-cover product (snow vs. no-snow) reprojected to 1 m and temporally averaged; plus the instantaneous satellite snow-cover extent (SCE) of the acquisition date.

Snow persistence.. The persistence channels summarise the recent snow history of each pixel. For a given acquisition date and temporal window $W \in \{15, 30, 60\}$ days, the binary snow-cover product (snow $= 1$, no-snow/cloud

= 0) is reprojected from its native 10 m resolution to 1 m by bilinear resampling and averaged over all available observations within the window, yielding a value in $[0, 1]$ that represents the fraction of recent days on which the pixel was snow-covered. The ablation study (Section 5.3) shows that these channels are among the most informative predictors.

Meteorological channels (extended configuration).. To test whether explicit meteorological forcing improves prediction, we build an extended 26-channel configuration that adds four scalar channels derived from the catchment automatic weather station (provided by IPE–CSIC): the mean air temperature over the 7, 15 and 30 days preceding the acquisition, and the precipitation accumulated since the 1st of October of the corresponding hydrological year. These are scalar (spatially uniform) per date: every pixel of a given acquisition shares the same value, so the channels encode the temporal meteorological context rather than spatial structure.

3.4. Temporal split

To obtain an honest estimate of predictive skill on *unseen conditions*, we use a strictly temporal train/validation/test split rather than a random pixel or tile split, which would leak spatial information between sets:

- **Training:** hydrological years 2021–2023 (15 dates, 4241 tiles).
- **Validation:** 2024 (7 dates, 431 tiles).
- **Test:** 2025 (5 dates, 220 tiles).

The test year 2025 was an exceptionally snowy season: the mean test HS (1.09 m) is 51% higher than the mean training HS (0.72 m). This distribution shift is deliberate – it stresses the extrapolation ability of the models and explains the moderate absolute R^2 values and the non-negligible seed-to-seed variance observed in Section 5. We regard robustness under this shift as a key, realistic requirement for operational snow mapping.

4. Methods

4.1. Overview and preprocessing

All models predict HS at 1 m resolution from the predictor stack described in Section 3.3. Each channel is normalised to a comparable range using fixed,

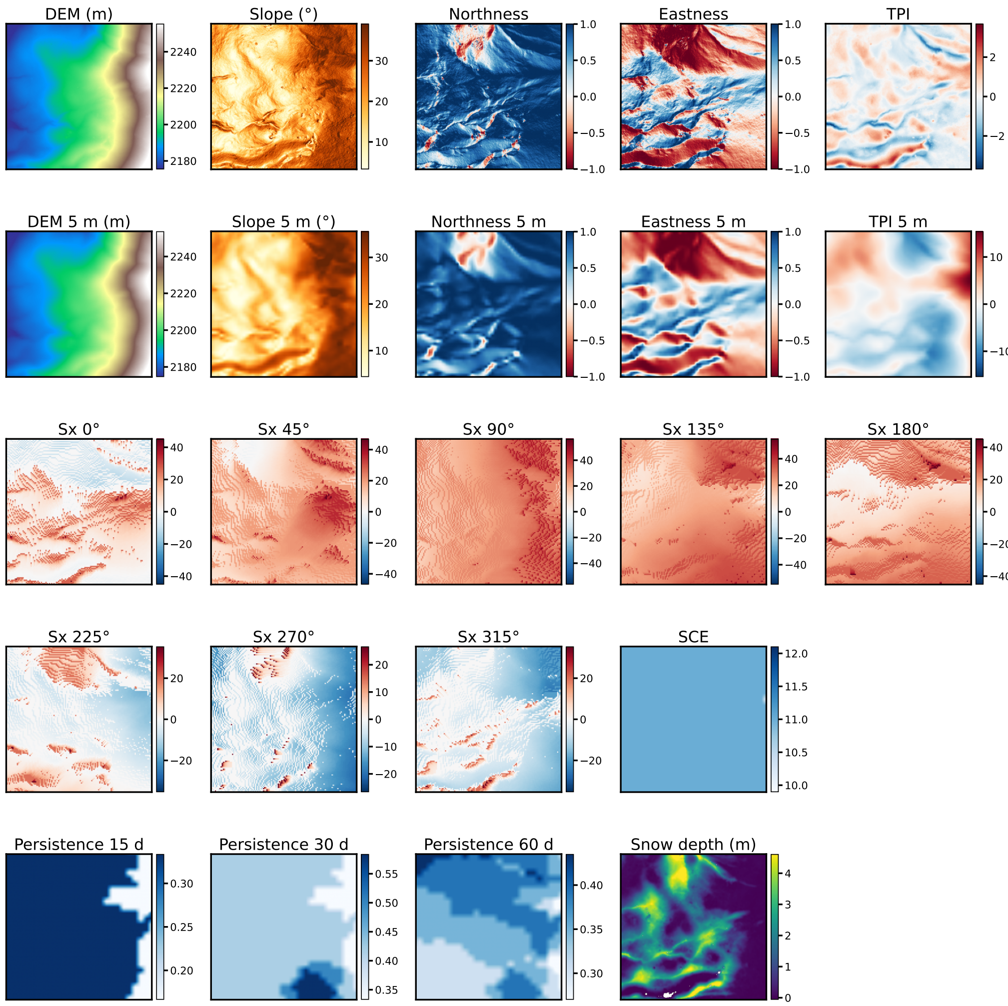

Figure 2: Example predictor stack for a single 256 × 256 m tile (2 February 2021). Rows, top to bottom: fine 1 m topographic derivatives (DEM, slope, northness, eastness, TPI); the corresponding coarse 5 m derivatives; the eight S_x wind-sheltering indices (0–315°); and the satellite snow-cover extent (SCE), the 15/30/60-day snow-persistence channels and the target LiDAR snow-depth map. Each panel is independently scaled; diverging fields are centred at zero.

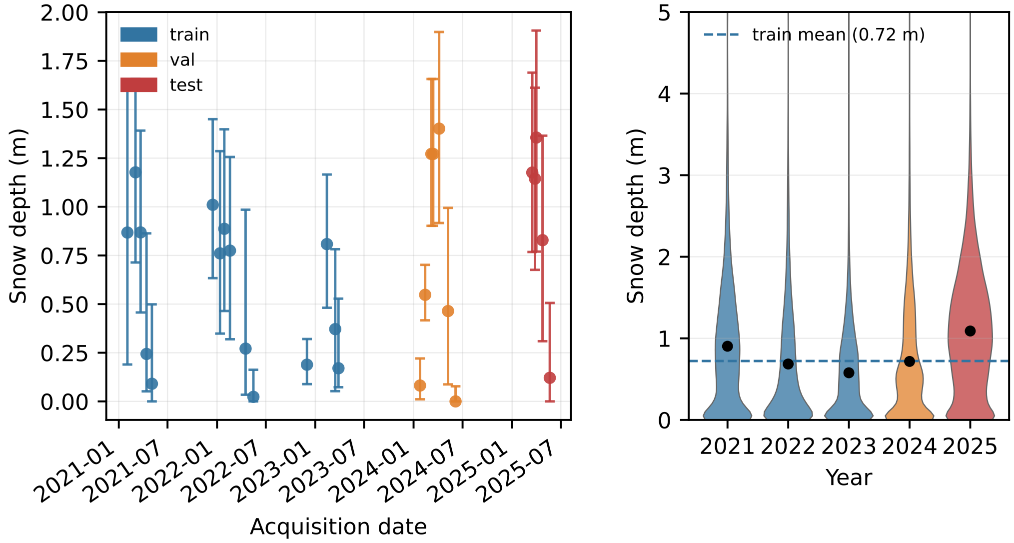

Figure 3: Temporal train/validation/test split and the corresponding distributions of LiDAR snow depth. (a) Per-date median snow depth (± interquartile range) along the acquisition timeline, coloured by split. (b) Per-year snow-depth distributions (violins); black dots mark the annual means and the dashed line the training mean. The 2025 test year is markedly snowier than the 2021–2023 training years, illustrating the distribution shift that the models must extrapolate to.

physically motivated scalings (e.g. elevation is standardised with a reference mean and standard deviation, slopes are divided by 90°, the TPI is clipped to $[-1, 1]$, the S_x indices are divided by 90°, persistence channels are already in $[0, 1]$, and SCE is binarised). Invalid sentinel values (-9999) and NaNs are replaced by zero after normalisation. Pixels without valid HS are masked out of the loss and of every metric. Crucially, normalisation is applied to the full 22-channel stack *before* any channel subset is selected, so that the ablation experiments (Section 5.3) do not alter the normalisation statistics of the remaining channels.

4.2. Models

Random Forest (baseline).. As a strong non-spatial baseline we train a Random Forest regressor [14] on the per-pixel feature vectors. Hyperparameters were optimised with Optuna [23] and the final configuration uses 300 trees, a maximum depth of 15, a minimum of one sample per leaf, `log2` feature subsampling and a minimum of two samples to split. The RF is trained on the union of the training and validation pixels (standard refit after hyperparameter optimisation) and predicts each pixel independently; spatial metrics are computed by reassembling the per-pixel predictions into their original tiles.

U-Net.. The U-Net [16] is a symmetric encoder–decoder with skip connections. Our implementation uses a base width of 48 feature maps and the standard four-level contracting/expanding path. It is trained end-to-end as a pixel-wise regressor.

ResUNet++.. ResUNet++ [18] augments the residual U-Net [17] with squeeze-and-excitation blocks in the encoder, an atrous spatial pyramid pooling (ASPP) bridge, and attention gates in the decoder. These components improve gradient flow and multi-scale context aggregation. We use a base width of 64 and Group Normalisation (8 groups) instead of Batch Normalisation, which we found essential for stable training under the small batch sizes imposed by high-resolution tiles and the heterogeneous, date-dependent statistics of the snowpack.

4.3. Spatial evaluation metrics

Pixel-wise metrics.. We report the coefficient of determination (R^2, equivalently the Nash–Sutcliffe efficiency), the mean absolute error (MAE), the root mean squared error (RMSE) and the bias, all computed over the valid test pixels.

SPAEF.. To assess the fidelity of the predicted *spatial pattern* we use the Spatial Efficiency metric [20], defined as

$$\mathrm{SPAEF} = 1 - \sqrt{(\alpha - 1)^2 + (\beta - 1)^2 + (\gamma - 1)^2}, \tag{1}$$

where the three components measure complementary aspects of pattern agreement:

$$\alpha = \rho(o, s), \qquad \beta = \frac{\sigma_s/\mu_s}{\sigma_o/\mu_o}, \qquad \gamma = \frac{\sum_{j=1}^{K} \min\left(h_j^o, h_j^s\right)}{\sum_{j=1}^{K} h_j^o}. \tag{2}$$

Here o and s are the observed and simulated fields, ρ is the Pearson correlation coefficient (α measures linear co-location of the pattern), β is the ratio of the coefficients of variation (it penalises over-/under-dispersed patterns), and γ is the intersection of the K-bin histograms h^o and h^s of the two fields (it measures the agreement of the value distributions). A perfect match yields SPAEF = 1; the metric is unbounded below. SPAEF is computed per tile and averaged over the test set.

MSPAEF.. Because snow-redistribution patterns operate at multiple spatial scales, we also report a multi-scale SPAEF (MSPAEF), obtained by evaluating the SPAEF components on successively coarsened versions of the fields and aggregating them. MSPAEF rewards models that reproduce structure across a range of scales rather than at the native resolution only, and is reported as a complementary indicator.

4.4. Hybrid spatial loss

Training a CNN with a pure pixel-wise MSE encourages regression to the mean and tends to produce over-smoothed maps. To explicitly reward spatial pattern fidelity, we train the CNNs with a hybrid loss that linearly combines the pixel-wise MSE with a spatial Pearson-correlation term:

$$\mathcal{L} = (1-\lambda)\underbrace{\mathrm{MSE}(o,s)}_{\text{pixel-wise}} + \lambda\underbrace{\big(1-\rho_{\text{spatial}}(o,s)\big)}_{\text{pattern}}, \tag{3}$$

where ρ_{spatial} is the Pearson correlation between the predicted and observed fields computed over the valid pixels of each tile, and $\lambda \in [0, 1]$ controls the trade-off between absolute accuracy and pattern agreement. With $\lambda = 0$ the loss reduces to plain MSE; increasing λ progressively prioritises the reproduction of the spatial pattern. The correlation term is differentiable and adds negligible computational cost. We characterise the effect of λ through a full sweep (Section 5.2).

4.5. Training and evaluation protocol

Hyperparameter optimisation.. The architecture and optimiser hyperparameters of each model (base width, learning rate, optimiser, batch size, weight decay, dropout, gradient clipping) were tuned with Optuna [23] on the train/validation split. The resulting ResUNet++ configuration uses the AdamW optimiser with a learning rate of 1.29×10^{-4}, batch size 8, dropout 0.077 and weight decay 1.24×10^{-4}; the U-Net uses Adam with a learning rate of 3.06×10^{-5} and batch size 16.

Training.. All CNN models are trained for 50 epochs without early stopping and without data augmentation, to keep the comparison controlled and reproducible. Because the 2025 test year lies outside the training distribution, the validation loss is an imperfect proxy for test performance. We therefore retain two checkpoints per run – the best-validation epoch and the last epoch – and report, for each architecture, the checkpoint selection that is most appropriate for it: ResUNet++ is evaluated at its best-validation checkpoint (which correlates well with test skill), whereas U-Net is evaluated at its last epoch, since its best-validation checkpoint yields degenerate test predictions. This asymmetry is itself an informative result about the stability of the two architectures (Section 6).

Reproducibility and seeds.. Every CNN experiment is repeated with three random seeds (42, 123, 7), with the seed fixed immediately before model construction. We report the mean and standard deviation across seeds for all metrics; the seed spread is an essential part of our results, as it reveals stability differences that single-run numbers would mask. The RF, being almost deterministic for this problem, is likewise evaluated over the three seeds for consistency.

Experiments.. Four experiments are reported in Section 5: (i) a head-to-head comparison of RF, U-Net and ResUNet++ on the 22-channel dataset; (ii) a sweep of the spatial-loss weight $\lambda \in \{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0\}$ for ResUNet++ (eight values $\times$ three seeds); (iii) a leave-one-group-out ablation of the predictor families (six configurations $\times$ three seeds); and (iv) an analysis of the 26-channel meteorological configuration.

5. Results

All results are reported on the temporally disjoint 2025 test set, averaged over three seeds (mean $\pm$ standard deviation) unless stated otherwise.

5.1. Model comparison: RF vs. U-Net vs. ResUNet++

Table 2 compares the three models on the 22-channel dataset. ResUNet++ trained with the optimal spatial loss ($\lambda = 0.4$, Section 5.2) attains the best scores on every metric: $R^2 = 0.208 \pm 0.027$ and SPAEF $= 0.297 \pm 0.018$. The Random Forest is a solid baseline in terms of R^2 (0.140 ± 0.004) and is remarkably stable, but its spatial pattern fidelity is poor (SPAEF $= 0.107$):

Table 2: Model comparison on the 22-channel dataset (2025 test set, mean ± std over seeds 42/123/7). U-Net is evaluated at its last epoch, ResUNet++ at its best-validation epoch (see Section 4.5). Best value per column in bold. SPAEF for RF is computed by reassembling per-pixel predictions into tiles.

Model	R^2↑	SPAEF↑	MAE↓	RMSE↓	Bias	R^2 std
Random Forest	0.140	0.107	0.515	0.675	−0.307	±0.004
U-Net (last epoch)	0.036	0.088	0.541	0.709	−0.223	±0.282
ResUNet++ (λ=0.4)	**0.208**	**0.297**	**0.477**	**0.648**	−0.282	±0.027

Table 3: Per-seed R^2 and SPAEF, illustrating the seed-to-seed (in)stability of each model.

	R^2			SPAEF		
Model	s42	s123	s7	s42	s123	s7
Random Forest	0.140	0.136	0.144	0.104	0.110	0.108
U-Net	0.362	−0.141	−0.112	0.346	−0.052	−0.030
ResUNet++ (λ=0.4)	0.186	0.200	0.239	0.294	0.281	0.317

operating per-pixel, it cannot reproduce the coherent drift structures of the snowpack.

The most striking result concerns U-Net. Its *mean* R^2 across seeds is only 0.036, with a standard deviation of ±0.282 – larger than the mean itself. A single fortunate run (seed 42) reaches $R^2 = 0.362$, the highest single-run value in the whole study, but the other two seeds collapse to negative R^2. The pattern-oriented SPAEF tells the same story (0.088 ± 0.216): U-Net is, on average, no better than the per-pixel RF and far less reliable than ResUNet++. This is precisely the kind of instability that a single-run, R^2-only evaluation would hide, and it motivates both our three-seed protocol and the use of SPAEF.

5.2. Spatial-loss weight sweep

Table 4 and Fig. 6 report the effect of the spatial-loss weight λ on ResUNet++. Both R^2 and SPAEF follow a clear concave trend: starting from pure MSE ($\lambda = 0$), adding a moderate amount of spatial correlation improves performance up to an optimum at $\lambda = 0.4$, after which over-weighting the pattern term (towards $\lambda = 1.0$, i.e. pure correlation) degrades both metrics. At the optimum, R^2 rises from 0.139 ± 0.111 ($\lambda = 0$) to 0.208 ± 0.027, and

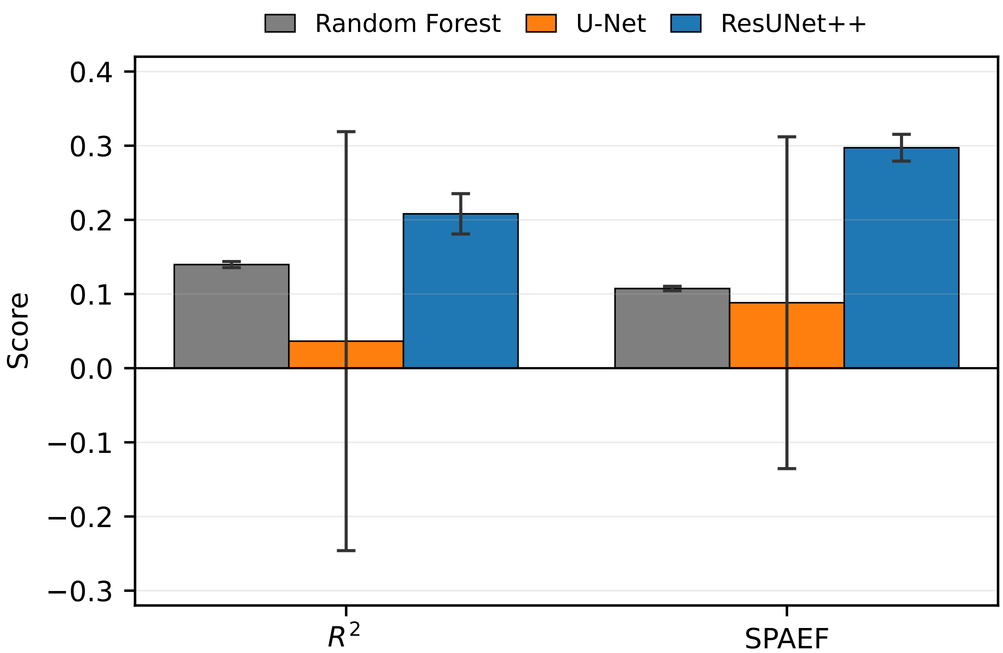

Figure 4: Model comparison on the 2025 test set. Bars show the mean over three seeds and error bars the standard deviation, for R^2 (left) and SPAEF (right). ResUNet++ (λ=0.4) dominates on both metrics with the smallest variance, while U-Net exhibits extreme seed-to-seed instability (its error bars cross zero).

SPAEF from 0.212 ± 0.123 to 0.297 ± 0.018 – the spatial loss not only improves the mean but also *halves* (in fact reduces by $\sim$80%) the seed-to-seed variance, making training markedly more reproducible.

5.3. Channel ablation

To quantify the contribution of each predictor family we perform a leave-one-group-out ablation on ResUNet++, removing one channel family at a time and retraining from scratch (three seeds each). Table 5 reports the resulting test scores; larger drops relative to the full model indicate more important channels.

Two families are critical. Removing the 5 m (coarse) topography collapses the model to $R^2 = -0.114$ and SPAEF = 0.076, and removing the snow persistence channels is equally damaging ($R^2 = -0.148$, SPAEF = 0.129). This confirms that meso-scale terrain context and recent snow history are fig05_prediction_maps.pdf

Figure 5: Qualitative comparison of predicted snow-depth maps for a representative 2025 test tile. From left to right: LiDAR ground truth, Random Forest, U-Net (last epoch) and ResUNet++ (λ=0.4). ResUNet++ best reproduces the wind-driven accumulation pattern, whereas RF produces a spatially incoherent field and U-Net over-smooths. **[PLACEHOLDER FIGURE]**

fig06_lambda_sweep.pdf

Figure 6: Effect of the spatial-loss weight λ on ResUNet++. Curves show the mean R^2 and SPAEF over three seeds with shaded ± 1 std bands. Both metrics peak at $\lambda = 0.4$; the shaded band also contracts around the optimum, indicating improved reproducibility. **[PLACEHOLDER FIGURE]**

Table 4: Spatial-loss weight sweep for ResUNet++ on the 22-channel dataset (mean ± std over seeds 42/123/7). The optimum is $\lambda = 0.4$ for both metrics, which also yields the lowest variance.

λ	R^2	R^2 std	SPAEF	SPAEF std
0.00	0.139	±0.111	0.212	±0.123
0.10	0.173	±0.025	0.267	±0.018
0.25	0.152	±0.079	0.272	±0.034
0.40	**0.208**	±0.027	**0.297**	±0.018
0.50	0.203	±0.019	0.288	±0.010
0.60	0.145	±0.045	0.235	±0.032
0.75	0.181	±0.076	0.278	±0.043
1.00	0.119	±0.035	0.194	±0.060

Table 5: Leave-one-group-out channel ablation for ResUNet++ (22-channel full model, mean ± std over three seeds). Rows are sorted by impact; the largest degradations (most important channels) are at the top.

Configuration	R^2	SPAEF
Full (22 channels)	0.132 ± 0.112	0.251 ± 0.051
− Snow persistence	-0.148 ± 0.113	0.129 ± 0.026
− Topography 5 m	-0.114 ± 0.119	0.076 ± 0.080
− S_x wind sheltering	0.089 ± 0.097	0.225 ± 0.044
− Topography 1 m	0.079 ± 0.092	0.220 ± 0.048
− SCE (snow-cover extent)	0.132 ± 0.027	0.242 ± 0.022

the backbone of the prediction. The S_x wind-sheltering channels and the 1 m topography are moderately important, while the instantaneous satellite snow-cover extent (SCE) is nearly redundant – its removal leaves R^2 essentially unchanged and even reduces variance, consistent with the persistence channels already encoding the relevant snow-cover information.

Note. The ablation is performed at $\lambda = 0$ (pure MSE) so that all configurations are compared under identical, loss-neutral conditions; the full-model baseline in Table 5 therefore corresponds to the $\lambda = 0$ row of Table 4 and is lower than the $\lambda = 0.4$ optimum used elsewhere.

fig07_ablation.pdf

Figure 7: Channel-family ablation for ResUNet++. Bars show the change in R^2 and SPAEF when each family is removed, relative to the full 22-channel model (mean $\pm$ std over three seeds). Snow persistence and 5 m topography are the most influential predictors; SCE is nearly redundant. **[PLACEHOLDER FIGURE]**

Table 6: Effect of adding meteorological channels. Best 22-channel ResUNet++ (λ=0.4) vs. best 26-channel configuration (λ=0.6); mean ± std over three seeds.

Configuration	R^2	SPAEF	MSPAEF	MAE	RMSE	R^2 std
22-ch (λ=0.4)	0.208	**0.297**	0.462	0.477	0.648	±**0.027**
26-ch (λ=0.6)	**0.213**	0.283	**0.463**	0.480	**0.646**	±0.054

5.4. Meteorological channels (extended configuration)

Finally, we test whether adding explicit meteorological forcing improves prediction. Table 6 compares the best 22-channel ResUNet++ ($\lambda = 0.4$) with the 26-channel configuration that adds the four temperature/precipitation channels; for the latter the best operating point of the λ sweep was $\lambda = 0.6$. The two configurations are statistically indistinguishable on the central metrics: the 26-channel model is marginally better on R^2 (0.213 vs. 0.208) and the 22-channel model is marginally better on SPAEF (0.297 vs. 0.283), with overlapping standard deviations on every metric. Moreover, the 22-channel model is *more stable* (R^2 std ±0.027 vs. ±0.054).

We conclude that, for this catchment and predictor set, the topographic and snow-history channels already capture most of the explainable variance, and the scalar (spatially uniform) meteorological channels add little complementary spatial information. We therefore retain the 22-channel configuration as our reference model and report the meteorological experiment as a complementary analysis.

6. Discussion

6.1. Pattern-aware metrics change the ranking

The headline finding of our study is methodological: *how* a high-resolution snow model is evaluated can change which model one would select. Judged by the single best U-Net run ($R^2 = 0.362$), one would declare U-Net the winner. Judged by the three-seed mean and by SPAEF, ResUNet++ is clearly superior and U-Net is barely competitive with a per-pixel Random Forest. The discrepancy arises because R^2 rewards getting the overall magnitude right and is dominated by the easy, smoothly varying part of the field, whereas SPAEF jointly penalises errors in spatial correlation, dispersion and value distribution – the components that encode the wind-driven drift structures of interest. For applications that depend on the *location* of deep and fig08_meteo_comparison.pdf

Figure 8: Comparison of the 22-channel and 26-channel (with meteorological forcing) ResUNet++ configurations across all metrics (mean ± std over three seeds). Differences fall within the seed variance, indicating no significant benefit from the added meteorological channels. **[PLACEHOLDER FIGURE]**

shallow snow (avalanche release zones, meltwater routing, ecological niches), SPAEF/MSPAEF are the operationally relevant criteria, and we recommend reporting them alongside pixel-wise metrics in future high-resolution snow studies.

6.2. Architecture stability matters as much as accuracy

ResUNet++ is not only more accurate on average but dramatically more *reproducible*: its R^2 standard deviation across seeds (± 0.027 at $\lambda = 0.4$) is an order of magnitude smaller than U-Net's (± 0.282). We attribute this to two design choices. First, the residual connections and attention/ASPP blocks of ResUNet++ provide a well-conditioned optimisation landscape in which the network can learn incrementally from an identity mapping, reducing sensitivity to initialisation. Second, Group Normalisation – unlike the Batch Normalisation used in our U-Net – is independent of batch statistics and is therefore robust to the small batch sizes and the heterogeneous, date-dependent snow distributions of our data. The fact that U-Net requires its *last*-epoch checkpoint (its best-validation checkpoint is degenerate) while ResUNet++ behaves well at best-validation is a further symptom of this instability. In an operational setting where a single model must be trained and trusted, this reproducibility is arguably as important as the marginal accuracy gain.

6.3. What drives the prediction?

The ablation identifies snow persistence and 5 m topography as the two indispensable predictor families, with the S_x wind-sheltering indices and 1 m topography as secondary contributors. This is physically coherent: recent snow history is a strong autoregressive prior on current depth, and meso-scale terrain governs the wind redistribution that creates the dominant accumulation patterns at Izas. The near-redundancy of the instantaneous satellite SCE is explained by the persistence channels, which are themselves temporal aggregates of the same snow-cover product and carry the relevant information in a less noisy form.

6.4. Why meteorological forcing did not help

Adding temperature and accumulated-precipitation channels produced no significant improvement. We see two reasons. First, the meteorological inputs are *scalar per date*: they shift the whole tile up or down but provide no spatial structure, so they can only help to the extent that the date-level snow magnitude is not already inferable from the snow-history channels – which it largely is. Second, the small number of distinct acquisition dates limits the amount of temporal signal the network can exploit before over-fitting. A spatially distributed meteorological forcing (e.g. gridded temperature, incoming radiation or modelled wind fields) might carry complementary spatial information; we regard this as a promising direction rather than a negative result about meteorology per se.

6.5. Limitations

Several limitations temper our conclusions. (i) The benchmark is a single, small catchment; transferability to other terrains and snow climates remains to be demonstrated. (ii) The test set is a single, exceptionally snowy year; while this is a deliberate and demanding extrapolation test, it also inflates the seed variance and limits the precision of the absolute scores. (iii) The number of LiDAR dates (27) is small by deep-learning standards, which constrains model capacity and amplifies initialisation sensitivity. (iv) The persistence channels treat clouds in the satellite snow-cover product as “no-snow”, which may bias the recent-history estimate during prolonged cloudy periods; a cloud-aware temporal gap-filling could refine this input. (v) MSPAEF is reported as a complementary indicator but its multi-scale aggregation could be standardised further. Addressing these points – multi-catchment training, more acquisition dates, distributed meteorological forcing and cloud-aware snow-history – defines our future work.

7. Conclusions

We presented a reproducible benchmark for high-resolution (1 m) snow-depth prediction in the Izas experimental catchment of the Central Spanish Pyrenees, built from a 27-date airborne LiDAR time series and a rich stack of topographic, wind-sheltering and snow-history predictors provided in collaboration with IPE–CSIC. Using a strictly temporal train/validation/test split that forces extrapolation to an exceptionally snowy year, we compared a Random Forest baseline with U-Net and ResUNet++ convolutional networks, and we evaluated all models with pattern-aware spatial metrics (SPAEF, MSPAEF) in addition to conventional pixel-wise scores.

Our main conclusions are:

- **ResUNet++ with a spatial loss is the best and most reliable model.** At the optimal spatial-loss weight $\lambda = 0.4$ it achieves $R^2 = 0.208 \pm 0.027$ and SPAEF $= 0.297 \pm 0.018$, outperforming both the Random Forest (SPAEF $= 0.107$) and a highly unstable U-Net (SPAEF $= 0.088 \pm 0.216$).

- **Pattern-aware evaluation is essential.** The SPAEF metric and the three-seed protocol expose a seed-to-seed instability of U-Net that a single-run, R^2-only assessment would completely miss. We recommend reporting SPAEF/MSPAEF and multi-seed statistics as standard practice in high-resolution snow mapping.

- **The hybrid spatial loss helps twice.** Blending pixel-wise MSE with a spatial-correlation term improves both accuracy and pattern fidelity and substantially reduces the variance across seeds, with a clear optimum at $\lambda = 0.4$.

- **Snow history and meso-scale topography are the key predictors,** whereas instantaneous satellite snow-cover extent is nearly redundant, and scalar meteorological forcing does not significantly improve the prediction for this catchment.

By coupling a stable, attention-augmented architecture with a pattern-oriented loss and evaluation, this work provides both a methodological template and a public benchmark for high-resolution snow-depth mapping. Future work will extend the approach to multiple catchments and snow climates, incorporate spatially distributed meteorological forcing and cloud-aware snow-history inputs, and assess transferability across years with contrasting snow conditions.

Code and data availability

The dataset, trained model weights and the code required to reproduce all experiments will be made available in a public repository (Zenodo / GitHub) upon acceptance.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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